

Pitches, Frequencies, and Tuning, Part 2

We've seen that instruments naturally produce series of harmonics. An instrument playing a note vibrates at a fundamental frequency that is the base pitch we hear. It will also generally vibrate at a set of higher frequencies, that ideally are integer multiples of the fundamental frequency. Different instruments will produce different amounts of the various harmonics, and that is a huge part of what we hear as the sound quality or timbre of an instrument. Even from the same instrument, the amount of the various harmonics will change for different fundamentals. An extreme example is a cello playing its lowest note, where the fundamental is almost completely missing but our brain figures out it *should* be there, so we hear it. For higher pitches, the cello does indeed produce the fundamental. For a string instrument, we would expect to get all integer multiples of the fundamental. For a flute that's open at one end, we would expect to get odd multiples.

1. Equal Tempering

We've also seen that mathematics doesn't let us have a "perfect" tuning system. What we hear as a perfect fifth are frequencies in the ratio of 3:2. If we play two notes that are slightly off that ratio, we will hear beating. It's impossible to set up a scale where everything we think of as a fifth is actually a perfect 3:2 ratio, so we have to make compromises. The simplest way to do this is to set the ratio of adjacent half steps to be a constant value. We have 12 half-steps in a scale (because 12 perfect 3:2 fifths brings us nearly back to where we started), and the spacing of an octave is exactly a factor of 2. That means the if the frequency ratio between two adjacent notes is r , then when we multiply r together 12 times, we should get exactly 2. Mathematically, we have $r^{12} = 2$ or $r = 2^{1/12} = 1.05946\dots$. The spacing of the notes in a scale is called *tempering*, and the uniform ratio is called *equal tempering*.

Let's practice starting with a fifth. Going from a C up to a G should be a 3:2 ratio, but what we'll actually get is a change of seven half-steps. Each half-step is a factor of r in frequency, so the actual ratio we'll get is $r^7 = 2^{7/12} = 1.4983$. This is pretty close to 1.5, although the difference can be audible.

If we look at other harmonics, what we hear (assuming we're starting at a C) is [C,C,G,C,E,G,b-flat,C...] for $\nu = [1, 2, 3, 4, 5, 6, 7, 8\dots]\nu_f$ where ν_f is the fundamental. A C to and E is a major third. If we read off the frequencies, the C to the next E is for $\nu = [4, 5]\nu_f$. Going up by a major third means the frequency ratio is $5\nu_f/4\nu_f = 1.25$. For our evenly-spaced notes, going from a C to an E is 4 half-steps, which gives a frequency ratio of $r^4 = 2^{4/12} = 2^{1/3} = 1.2599\dots$. This is nearly a percent off, and if you listen carefully, most people can hear the difference. We'll listen to this in class, but you can play around yourself if you like with the python code "play_tempered_chord.py" I have posted on github.

While we're here, let's also look at Pythagorean tuning, where we try to keep our fifths exactly perfect. If I go from C to G then the frequency ratio is exactly 1.5. Going from G to D gives us 1.5 again, or 9/4. We've gone up by just over an octave, so if we bring the D down by an octave, we divide its frequency by 2. That puts the frequency ratio between a C and a D at

$9/4/2 = 9/8 = 1.125$. A D to an E will be another factor of 1.125, for a total frequency ratio of $81/64 = 1.125^2 = 1.265625$. This is about 50% further off than the equal tempering from the desired value of 1.25. We made our fifths perfect, but at the expense of our thirds!

Example: A fourth should have a ratio of 4:3 because going from a G up to a C comes from 3rd and 4th harmonics. If you count on a keyboard, this is 5 half-steps, so the ratio we'll usually get is $r^5 = 2^{5/12} = 1.3348$. This is pretty close, so fourths sound not bad in equal tempering.

2. Pitch Notation and Larger Intervals

If we want to talk about larger intervals, or even just specific notes, it's convenient to have a system for labelling the notes. The standard notation is a letter for the usual note name, followed by a number telling us which octave a note is in. By convention, a middle C is a C4, mainly because it's the 4th octave on a standard piano. Confusingly, the numbers cycle at C and not A, so a B3 is a half-step below a C4. Today, the reference pitch widely used is that A4 (so the A *above* middle C) is at 440 Hz. This is not set in stone, and historically it has varied widely. Baroque instruments were tuned on average a full half-step lower. In the mid 1800's, the French set A4 to 435 Hz, and it wasn't until 1939 that 440 Hz was adopted. Even today, some orchestras use different tunings. Both the New York Philharmonic and Boston Symphony Orchestra tune to 442 Hz. An interesting recap of the pitch of A4 over the years is [here](#).

Now we can more easily talk about the frequencies of different notes, and larger intervals. The lowest note on a piano is an A 4 octaves below A4, so A0. What is the frequency of an A0? Well, we drop by a factor of 2 every time we go down an octave, so going down 4 octaves means we drop 4 factors of 2, or $2^4 = 16$. The frequency of an A0 is then $440/16 = 27.5$ Hz. This is getting close to the limit of human hearing at 20 Hz. You could only add a few notes to a piano and still have it sound like a note! How about middle C, which is C4? Remember, A3 is the A immediately below middle C, and it's 3 half-steps away (A to B-flat to B to C). An A3 is at $440/2$ or 220 Hz. A middle C is then $r^3 = 2^{3/12} = 2^{1/4} = 1.1892...$ times higher, or 261.6 Hz. Going from a C4 to a G4 is a fifth, because G4 is higher than C4. Going from G4 to C5 is a fourth, because C5 is the C immediately higher than a G4. I won't stress this notation, but it is convenient. It's also faster to just say "A1" for example, instead of "The A a little over two octaves below middle C".

3. Frequencies to Intervals, and Logarithms

We've seen how to go from notes to frequencies, and pairs of notes to frequency ratios. Sometimes we want to do the reverse. Given the frequency of a sound, what note does it correspond to? Given a pair of frequencies, what is the interval between them?

To start, let's try to find how many half-steps a frequency is away from A4/440 Hz. If I told you I was 17 half-steps down from A4, then the frequency has to be $440/r^{17} = 164.8$ Hz. More generally, the frequency of a note n half-steps from A4 is

$$\nu = 440 \times r^n \text{ Hz} \tag{1}$$

How would we go the opposite direction? Given a reference frequency ν_0 (likely 440 Hz, but you could pick anything) and some other frequency ν , how many half-steps are between them? Well, if we knew there are n half-steps, then we know that

$$\nu = \nu_0 r^n \quad (2)$$

Before we knew n but wanted to know ν . Now we know ν and want to know n . We can do this by taking *logarithms* or logs. In case you haven't used one in the last week, if $c = a^b$, then the log base a of c is b . As an example, 8 is 2^3 , so the log base-2 of 8 is 3. Logs make solving Equation 2 easy because if I know the log of c , then the log of c^d is just $d \log(c)$. Also, the log of xy is the log of x plus the log of y . If we take the log of Equation 2, we get:

$$\log \nu = \log \nu_0 + n \log r \quad (3)$$

$$n \log r = \log \nu - \log \nu_0 \quad (4)$$

$$n = \frac{\log \nu - \log \nu_0}{\log r} \quad (5)$$

We can use any base for Equation 5, but base 2 is particularly easy. We know that $r = 2^{1/12}$ so $\log_2 r = 1/12$ where the notation $\log_2$ means we are taking the log base-2 of a number. We can use that to write an even easier version of Equation 5, because $1/(1/12)=12$, and we get:

$$n = 12 (\log_2 \nu - \log_2 \nu_0) \quad (6)$$

$$n = 12 \log_2 \frac{\nu}{\nu_0} \quad (7)$$

where we used the fact that $\log a - \log b = \log a/b$ in the final line. A few different numbers are commonly used for the base of logarithms. The most common is $e = 2.718281728459045...$ which is what nature “wants” to use as a base. If you want to use Equation 7, make sure you specify base 2 to whatever you're using. For example, using python/numpy, `np.log(x)` gives you the log base- e , so you would need to use `np.log2(x)`.

Let's use Equation 7 to find the lowest pitch a human can hear. The bottom end of hearing is 20 Hz - how many half-steps below A0 at 27.5 Hz is that? We set $\nu_0 = 27.5$, $\nu = 20$ and plug in to get:

$$n = 12 \log_2 20/27.5 = -5.5 \quad (8)$$

The minus sign means we go *down* 5.5 half-steps from an A0. If you count down from an A by five half-steps, you end up at an E, so the lowest note humans can generally hear is an E0¹.

4. Summary

We've looked at how to convert between note names and frequency using equal tempering, and seen how far off different temperings can be. We've also seen how to convert frequency ratios to note spacings, in units of half-steps. I don't expect you to derive anything using the details of logs (at least for pitch/frequency), but you should be able to convert notes to frequencies and frequencies to notes.

¹Technically, it's a quarter-step below E0. You can't play that in standard tuning, but Angine de Poitrine could!